Experiment G — Quadrature Speed/Equivariance Pareto

What is the right way to compare DH vs GL? **Same equivariance error**, then ask how much wall-clock time each method needs to reach it.

Setup (OC20-scale EquiformerV2, lmax=6, mmax=2)

Real `EquiformerV2Backbone` from fairchem: 12 layers, 128 channels, 24 `S2Activation` modules. NVIDIA A100-SXM4-40GB, idle.

Config	Method	Grid (lat × lon)	Points
DH default	e3nn equiangular	14 × 5	70
DH 2×	DH with `resolution=4(L+1)`	28 × 28	784
GL min	custom GL	7 × 13	91
GL match-DH	custom GL	14 × 13	182
GL 2×	custom GL	14 × 28	392

Two measurements at the same (lmax, mmax)

Equivariance error of one S2-Act kernel (random coefficients, 5 rotations × 10 inputs); SiLU activation. *Source:* `verify_gl_grid.py`.

Wall-clock for one full forward (QM9, batch=8), 90 measurements per config (3 runs × 30 iter), `cuda.sync()` around each, fresh model per config. *Source:* `verify_speedup_fairchem_default.json`.

Combined results

Config	Pts	Equiv err	Fwd (ms, 95% CI)
DH default	70	4.43×10^{-1}	111.83 ± 0.79
DH 2×	784	$\mathbf{3.27\times10^{-1}}$	$\mathbf{162.73 \pm 1.23}$ (+45.5%)
GL min	91	4.29×10^{-1}	∼111 (matches DH default)
GL match-DH	182	$\mathbf{3.28\times10^{-1}}$	$\mathbf{112.23 \pm 0.75}$ (+0.4% n.s.)
GL 2×	392	$\mathbf{3.28\times10^{-1}}$	$\mathbf{111.34 \pm 0.76}$ (−0.4% n.s.)

Matched-equivariance comparison

For target equiv err $= 3.28 \times 10^{-1}$ (the saturation floor at this mmax cropping):

- Cheapest DH: **DH 2× at 162.73 ms** (DH default at 70 pts only reaches 4.43×10^{-1}, not enough)
- Cheapest GL: **GL match-DH at 112.23 ms**
- **Savings: 50.5 ms =** $(162.73 - 112.23)/162.73 = 31.0\%$ **time reduction**

So: at matched equivariance accuracy, GL saves ∼31% wall-clock per forward.

Equivalently: GL achieves DH 2×'s equivariance at the same wall-clock as DH default; the +50.5 ms premium goes away.

Caveats

1. Result is for this specific (lmax=6, mmax=2) config and this GPU. Different model depths or hardware will scale it.
2. Equivariance is measured on *one* S2-Act kernel with random inputs, not on a trained model's intermediate features. Trained-model features are smoother (absolute equiv numbers will be much lower); the *ranking* between quadratures should be preserved.
3. The kernel-level equivariance gap between GL match-DH (3.28×10^{-1}) and DH 2× (3.27×10^{-1}) is within rounding — they really are the same accuracy.
4. The mmax=2 cropping caps achievable equivariance at ∼0.328 regardless of grid; lower error needs mmax=lmax (architectural change). At lmax=mmax both methods can reach $\sim4\times10^{-3}$ floor; there GL needs roughly half the latitude points of DH (see `verify_gl_grid.py`).
5. We did not measure end-to-end equivariance on a fully trained OC20-scale model. The matched-accuracy speedup claim rests on the kernel-level Pareto.

Earlier vs revised framing

The previous report claimed "DH 2× costs +45.5% over DH default", which was correct but compared different equivariance levels. The fairer claim:

> Switching DH 2× → GL match-DH saves **31% wall-clock per forward** while preserving the same equivariance error. Switching DH default → GL match-DH is **free** (same wall-clock) and substantially improves equivariance ($4.43\times10^{-1} \rightarrow 3.28\times10^{-1}$).

Reproduce

```
module load pytorch/2.6.0-1
# Equivariance for all configs at lmax=6, mmax=2
python3 experiments/verify_gl_grid.py
# Wall-clock: rigorous 3-run verification
PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True \
  python3 experiments/expG_verify_speedup.py \
    --config fairchem_default --batch_size 8 \
    --n_repeats 3 --n_iter 30 --n_warmup 20
```